

Irreversibility of extreme precipitation intensity in global monsoon areas under multiple carbon neutrality scenarios

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ABSTRACT

The Global Monsoon Areas (GMAs), home to over half of the world's population, face escalating socio-economic risks from extreme precipitation events intensified by rising atmospheric carbon dioxide (CO₂). While previous studies have examined the irreversibility of the climate system following carbon neutrality, most have focused on single carbon neutrality scenarios with limited attention to these vulnerable areas. This study assesses the irreversibility of extreme precipitation intensity across seven GMA sub-regions under eight future scenarios, incorporating four carbon neutrality targets and two reduction rates, using simulations from a state-of-the-art climate model. Our results reveal that extreme precipitation intensity exhibits irreversible behavior in response to carbon neutrality forcing, failing to return to its initial level even when atmospheric CO₂ is reduced. This irreversibility is particularly pronounced when carbon neutrality timing is delayed, and the emission reduction rate is slow. Moreover, the irreversible response is nonlinear to the magnitude of carbon forcing, leading to distinct regional vulnerabilities, with some areas experiencing sharp increases in irreversibility by even small delays in reaching carbon neutrality. This region-specific behavior is largely attributed to increases in mean and variability of precipitation linked to irreversible El Niño-like warming and interhemispheric differential warming. Moisture budget analysis further shows that the intensified precipitation arises from the relative influence of thermodynamic (moisture flux) and dynamic (wind) drivers across regions. These findings highlight the urgency of rapid policy implementation in vulnerable regions and can provide a scientific basis for developing regional adaptation strategies to mitigate growing extreme precipitation risks.

1. Introduction

Earth's climate has always been dynamic, but the current rate of global warming is unprecedented, largely driven by the sharp increase in atmospheric carbon dioxide (CO₂) concentrations from human activities (Joos and Spahni, 2008; Quéré et al., 2018). Since industrialization, CO₂ levels have continued to increase and surpassed 400 ppm, disrupting Earth's radiative balance and causing approximately 1 °C of global mean surface temperature rise compared to pre-industrial levels (IPCC, 2023). These disruptions have the potential to irreversibly alter the climate system, leading to destructive impacts on biogeochemical and hydrological cycles, ecosystems, and biodiversity (IPCC, 2021; Santana-Falcón et al., 2023; Zitoun et al., 2024). The Intergovernmental Panel on Climate Change (IPCC) warns that crossing climate tipping points could result in irreversible changes, possibly triggering additional global warming that exceeds 2 °C above pre-industrial temperature levels

(McKay et al., 2022). For this reason, the Paris Agreement (COP21) set a goal to limit warming caused by human activities to below 2 °C, necessitating rapid and sustained reductions in CO₂ emissions and, eventually, net-negative emissions (IPCC, 2018; Kriegler et al., 2018).

Carbon dioxide removal (CDR) strategies have thus emerged as a key component of long-term climate mitigation (Keller et al., 2018; Vaughan and Lenton, 2011). Advanced climate models have been used to evaluate the potential effects of CDR, offering insights into mitigation pathways and adaptive responses (Solomon et al., 2009; Zickfeld et al., 2013). The response of the climate system under CDR scenarios is often characterized by hysteresis effects and irreversible changes, significantly affecting extreme weather and climate events such as droughts, floods, and disruptions to the hydrological cycle (Chadwick et al., 2013; Hwang et al., 2024; Im et al., 2024; Jo et al., 2022; Kim et al., 2022). Notably, irreversible patterns of surface ocean temperature changes induced during CDR transitions, such as El Niño-like warming and

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interhemispheric differential warming, can lead to asymmetric and regionally persistent changes in precipitation responses (Kim et al., 2022; Kug et al., 2022).

The Global Monsoon Areas (GMAs), comprising East/South Asia, North/South Africa, Central/South America, and Australia, are a critical component of the global climate system and sustain over half of the world's population. Climate change-induced shifts in extreme precipitation within GMAs pose significant socio-economic and ecological challenges (Wang et al., 2021; Wang and Ding, 2008). Recent studies highlight the complexity of monsoon responses to CO₂ changes by CDR and the potential irreversibility in monsoon dynamics. For example, Paik et al. (2023) showed that the global monsoon domain expands during increase phases of CO₂ and contracts during decreasing phases, exhibiting hysteresis where monsoon boundaries may not revert to their original states even after CO₂ levels decline. Similarly, Mondal et al. (2023) found that extreme precipitation changes were more pronounced during the CO₂ reduction phases than during increase phases, with approximately 69 % of global land areas, particularly in monsoon regions, experiencing irreversible alterations in precipitation extremes.

Intensified extreme precipitation events can impose severe socio-economic consequences across the Global Monsoon Areas. Persistent Rx1day extremes heighten flood risks, damage critical infrastructure, disrupt transportation networks, and cause crop failures and loss of livelihoods (Chuphal et al., 2025; Otto et al., 2023). These impacts are particularly acute in monsoon-dependent regions of Asia, Africa, and South America, where large populations rely on seasonal rainfall for agriculture and water resources (IPCC, 2021; Winsemius et al., 2016). In addition, irreversible increases in hydrological extremes under delayed carbon neutrality could exacerbate urban flooding, waterborne disease outbreaks, and forced displacement, posing long-term challenges for sustainable development and disaster risk reduction (Hirabayashi et al., 2013; IPCC, 2023).

Despite the insights from the previous works, most studies have employed a single CO₂ ramp-up and ramp-down scenarios, typically involving a fourfold increase in CO₂ followed by a symmetrical return to pre-industrial levels (Kim et al., 2022; Liu et al., 2023; Oh et al., 2022; Z. Xu et al., 2024). This approach offers limited applicability in evaluating the impacts of various carbon neutrality timelines with different CO₂ reduction rates. Moreover, previous research has primarily focused on the irreversibility of seasonal precipitation changes, with limited emphasis on extreme precipitation events despite their potential to cause severe socioeconomic damage (Kug et al., 2022; Shi et al., 2021; Su et al., 2025; Wu et al., 2010; Xu et al., 2022). Additionally, analyses of region-specific vulnerabilities related to extreme precipitation across the GMAs are limited.

To address these gaps, this study employs eight distinct carbon neutrality scenarios that combine four CO₂ peak concentrations (440, 470, 570, and 1 145 ppm) with two removal rates (1 % and 2 % per year). This expanded framework enables a systematic assessment of how both the timing of carbon neutrality and the pace of CO₂ decline influence the magnitude and regional patterns of precipitation irreversibility (see the Method section for further details). Specifically, our design addresses two key scientific questions: (1) How sensitive is the irreversibility of extreme precipitation to the timing of achieving carbon neutrality? and (2) To what extent can faster CO₂ removal mitigate the degree of irreversibility across global monsoon regions?

To answer these questions, we assess how variations in the timing of carbon neutrality and the rate of CO₂ reduction influence the extent of irreversibility across the GMAs. Second, we analyze region-specific vulnerabilities associated with extreme precipitation, identifying areas most susceptible to irreversible climatic shifts. Finally, we explore the physical mechanisms underlying the irreversible changes to enhance our understanding of how extreme precipitation may persist or intensify even under future climate mitigation efforts.

2. Data and methods

2.1. Model experiment design

The model used to perform carbon neutrality simulations is UKESM1, the successor to the HadGEM2-ES model. The model was developed by the UK Met Office in collaboration with the Natural Environment Research Council (NERC) (Sellar et al., 2019), and the model version used in this study is the one operated by the National Institute of Meteorological Sciences (NIMS) in Korea. Thus, it is referred to as NIMS_UKESM throughout this study. The physical climate component is based on HadGEM3-GC3.1 (Senior et al., 2020), and it is coupled with terrestrial and ocean biogeochemical cycles. The model has a horizontal resolution of 1.875° in longitude and 1.25° in latitude, with 34 vertical levels. To address uncertainties related to internal model variability, three ensemble runs with different initial conditions were conducted.

All analyses in this study are based on the ensemble mean of three initial-condition members unless otherwise noted. For each ensemble member, key indices such as Rx1day and the monsoon precipitation index (MPI) were first calculated individually and then averaged to obtain the ensemble mean, representing the externally forced climate response. This ensemble approach follows standard CMIP practices, allowing the suppression of internal variability while highlighting robust, forced responses to changes in CO₂ forcing. All ensemble experiments were branched from a fully equilibrated model spin-up following the procedure described by Yool et al. (2020). This spin-up protocol involves component-wise equilibration of the ocean, land, and atmospheric chemistry modules, ensuring minimal model drift prior to the transient simulations.

The carbon neutrality scenarios designed by the NIMS, Korea, to force the model are based on four different timings of carbon neutrality, each combined with two carbon reduction rates (Fig. 1). Atmospheric CO₂ concentrations increase by 1 % per year from pre-industrial levels until they reach the specified carbon neutrality thresholds, A (440 ppm), B (470 ppm), C (570 ppm), and D (1 145 ppm), after which they decrease by either 1 % or 2 % per year. The carbon neutrality scenario with the threshold D follows the Carbon Dioxide Removal Model Intercomparison Project (CDRMIP) protocol (Keller et al., 2018), in which CO₂ concentrations increase by 1 % per year until quadrupling the pre-industrial level over 140 years and decrease at the same rate over the next 140 years, ultimately returning to the initial concentration. Scenarios A, B, and C correspond to the CO₂ levels at which carbon neutrality is reached in the Coupled Model Intercomparison Project Phase 6 (CMIP6) pathways, SSP1-1.9, SSP1-2.6, and SSP5-3.4OS, respectively. After CO₂ concentrations return to their initial levels, they are held constant. These experiments are labeled as A1 %, B1 %, C1 %, D1 %, A2 %, B2 %, C2 %, and D2 %, corresponding to the four neutrality points combined with 1 % or 2 % annual reduction rates.

To quantify irreversibility in extreme precipitation intensity over the GMAs, we analyze seven sub-regions (Fig. 3a): East Asia (EA; 110°E–145°E, 30°N–55°N), North Africa (NAF; 10°W–40°E, 5°N–22°N), South Asia (SA; 62°E–110°E, 5°N–28°N), Central America (CAM; 70°W–120°W, 4°N–28°N), South Africa (SAF; 10°E–40°E, 4°S–25°S), Australia (Aus; 120°E–150°E, 8°S–26°S), and South America (SAM; 100°W–145°W, 6°S–30°S). For analysis period, the first 30 year period of CO₂ increase phase is labeled as P_Up, the 30 year period when CO₂ reached its peak is labeled as P_Peak, and the 30 year ramp-down periods when CO₂ levels decrease back to the same level as P_Up is labeled as P_Down. The choice of a 30-year averaging window follows the World Meteorological Organization (WMO) standard for defining climate normals, which recommends a minimum of 30 years to minimize interannual and decadal variability and to robustly capture long-term climate signals (WMO, 2017). Irreversibility is quantified as the difference in extreme precipitation index between P_Down and P_Up. Note that the same P_Down period is used for both the 1 % and 2 % annual CO₂ reduction experiments.

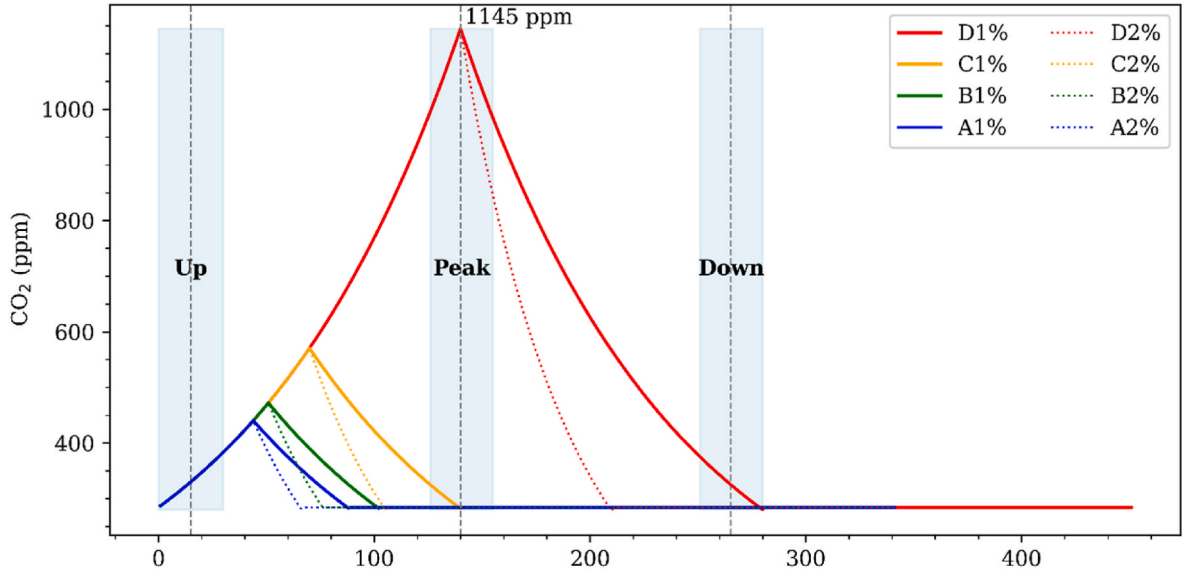


Fig. 1. Prescribed CO₂ concentrations for the A1 %, B1 %, C1 %, D1 %, A2 %, B2 %, C2 %, and D2 % carbon neutrality experiments (unit: ppm). The x-axis shows the simulation years, and the y-axis shows CO₂ concentration. Solid lines represent experiments with a 1 % annual CO₂ reduction rate, while dotted lines represent experiments with a 2 % annual reduction rate. The Up period denotes the 30-year phase of CO₂ increase (P_{Up}), Peak marks the 30-year period when concentrations remain at their maximum (P_{Peak}), and Down refers to the subsequent decline phase until concentrations return to the same level as in the Up period (P_{Down}).

2.2. Global monsoon areas

Monsoon climates are characterized by a strong contrast between wet summers and dry winters, along with a seasonal reversal in prevailing winds (Trenberth et al., 2000; Wang & Ding, 2006; Zhou et al., 2008). To objectively delineate the monsoon's domains, a previously defined Monsoon Precipitation Index (MPI) is used in this study, which is defined by the ratio of the annual range over the annual mean precipitation (Wang and Ding, 2008).

$$MPI = \frac{PR_{summer} - PR_{winter}}{PR_{Annual}} \quad (1)$$

where, PR_{summer} , PR_{winter} , and PR_{Annual} represent the mean precipitation during the summer, winter, and annual, respectively.

2.3. The index for extreme precipitation intensity

To assess chances in extreme precipitation intensity, we employ indices from the World Meteorological Organization's Expert Team on Climate Change Detection and Indices (ETCCDI), which comprises 11 precipitation-related indices and 16 temperature related indices (Zhang et al., 2011). This study focuses on the Rx1day index, which measures the annual maximum 1-day precipitation total, a widely used metric for characterizing extreme precipitation intensity.

Mathematically, Rx1day is expressed as:

$$Rx1day_j = \max(RR_{ij}) \quad (2)$$

where RR_{ij} is the daily precipitation amount on day i in year j . The maximum 1-day value for year j are $Rx1day_j$.

2.4. Moisture budget analysis

Precipitation can be expressed as the sum of evaporation and moisture convergence, enabling moisture budget analysis a key tool for understanding regional precipitation changes under climate forcing (Park et al., 2015a,b; Pomposi et al., 2015). To examine the precipitation changes in seven sub-regions of GMAs, the moisture budget was decomposed into dynamic (δDy), thermodynamic ($\delta ThDy$), eddy ($\delta Eddy$), and residual terms, and the precipitation difference between

P_{Down} and P_{Up} are analyzed for each moisture budget term, which is represented as follows:

$$\delta(Pr - Evap) = \delta ThDy + \delta Dy + \delta Eddy + Res. \quad (3)$$

$$\delta ThDy = - \int_0^{P_s} \nabla \cdot (\vec{U}_{p-up} \delta q) dp \quad (4)$$

$$\delta Dy = - \int_0^{P_s} \nabla \cdot (\delta \vec{U} q_{p-up}) dp \quad (5)$$

$$\delta Eddy = - \int_0^{P_s} \nabla \cdot (\delta (\vec{U}' q')) dp \quad (6)$$

where, Pr is precipitation, $Evap$ is evaporation from the surface, $\vec{U}$ is the horizontal wind vector, q is specific humidity, and p is pressure. δ represents the difference between the two period, P_{Down} and P_{Up}. The thermodynamic, dynamic and eddy terms are vertically integrated using a mass-weighted mean from surface pressure $P_s = 1\,000$ hpa to upper-level pressure $P_t = 200$ hpa.

2.5. Multiple linear regression model for extreme precipitation

To identify the large-scale climatic drivers of extreme precipitation responses to carbon neutrality forcing, we implemented a multiple linear regression model that relates regional precipitation changes across the GMAs to patterns of sea surface temperature (SST) changes. The regression is formulated as:

$$\Delta P = a \cdot \Delta SST_{NH-SH} + b \cdot \Delta SST_{EP-WP} + c \cdot \Delta SST_{IO} + d \cdot \Delta SST_{NA-SA} + d \quad (7)$$

where, ΔP represents the change in precipitation between P_{Down} and P_{Up}, ΔSST_{NH-SH} denotes the change in the interhemispheric SST gradient between the Northern Hemisphere and Southern Hemisphere, which controls ITCZ position and monsoon rainfall distribution (Chiang et al., 2008; He et al., 2022). ΔSST_{EP-WP} represents the change in the SST gradient between the Eastern Pacific and Western Pacific, reflecting

ENSO-like variability that strongly modulates East Asian monsoon extremes (Kug et al., 2022; Liu et al., 2023). ΔSST_{IO} indicates the change in the SST over the Indian Ocean (IO), a key driver of South and East Asian monsoon variability via enhanced moisture transport (Xu et al., 2019; Zhou et al., 2008). Finally, ΔSST_{NA-SA} corresponds to the change in Atlantic Meridional Mode (AMM), defined by the SST gradient between the North Atlantic and South Atlantic, which influence precipitation variability in South America and Africa through cross-basin teleconnections (Chiang and Vimont, 2004; Kushnir et al., 2002; Xie and Carton, 2004).

3. Results

3.1. Response of extreme precipitation intensity to multiple carbon neutrality scenarios across global monsoon areas

The GMAs exhibit pronounced seasonal variability in precipitation, as indicated by the MPI (Fig. 2a). This metric identifies monsoon regions that are highly sensitive to shifts in climate dynamics and provides a basis for evaluating the response of extreme precipitation to various carbon neutrality scenarios.

Extreme precipitation in this study is quantified using the annual maximum 1-day precipitation index (Rx1day). The model used in this study reproduces well the overall spatial pattern of Rx1day, although it underestimates its magnitude compared with the reanalysis data (Fig. S1). Fig. 2b–h presents the temporal evolution of Rx1day across the seven GMA sub-regions in response to various carbon neutrality forcing

experiments. Consistent with previous findings (Turner and Slingo, 2009), Rx1day in all monsoon regions intensifies with increasing atmospheric CO_2 , presumably due to the associated rise in atmospheric temperatures and specific humidity. Rx1day reaches its peak during or shortly after the carbon neutrality phase (P_Peak), but fails to fully return to its initial levels during the ramp-up phase (P_Up), even after CO_2 concentrations decline during the ramp-down phase (P_Down). In the D1 % scenario, for example, the difference in Rx1day between P_Up and P_Down ranges from approximately 1.5 to 6.8 mm/day across regions. This corresponds to a 5–12 % increase in P_Down relative to P_Up. This asymmetric response, defined in this study as irreversible, is particularly pronounced in scenarios with delayed carbon neutrality (Fig. S2). These results suggest that extreme precipitation does not simply reverse following CO_2 reductions, indicating the heightened risk of irreversible hydrological extremes as the achievement of carbon neutrality is delayed.

Scenarios with a 2 % annual CO_2 reduction show greater resilience compared to those with a 1 % reduction rate. While even delayed carbon neutrality scenarios such as D2 % still show increased Rx1day during the P_Down phase relative to P_Up, the faster drawdown of CO_2 can lead to more effective mitigation of Rx1day irreversibility. This suggests that more rapid CO_2 removal strategies may prevent long-term hydroclimate disruptions.

To quantify the irreversible responses of extreme precipitation intensity in different carbon neutrality scenarios, irreversibility was represented as the percentage change in Rx1day from P_Up to P_Down (Fig. 3). This represents the extent to which extreme precipitation fails

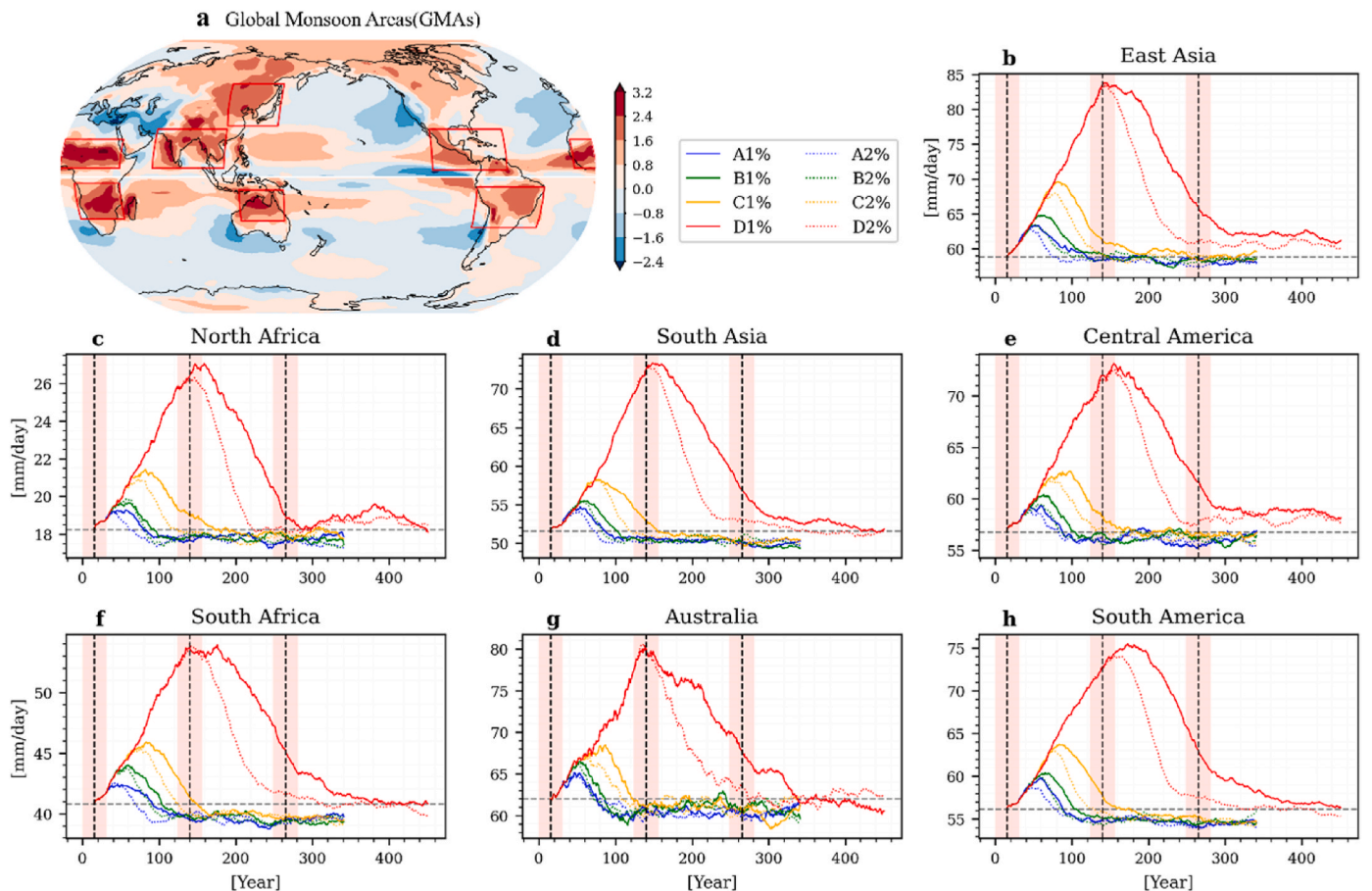


Fig. 2. (a) Spatial pattern of MPI, with red indicating above-average and blue indicating below-average summer-minus-winter monsoon precipitation relative to the annual mean climatology. (b–h) Time series of annual maximum 1-day precipitation (Rx1day) for each GMA under various carbon neutrality experiments. Blue, green, orange and red lines represent experiments A1 %, B1 %, C1 %, D1 %, A2 %, B2 %, C2 %, and D2 %, corresponding to various combinations of carbon neutrality timing and CO_2 reduction rates, with dotted lines denoting scenarios of 2 % per year reduction rate. Reddish vertical lines indicate the 30 years of CO_2 ramp-up (P_Up), peak (P_Peak), and ramp-down (P_Down) phases for D1 % and D2 %. All lines represent annual means smoothed using a 30-year running mean.

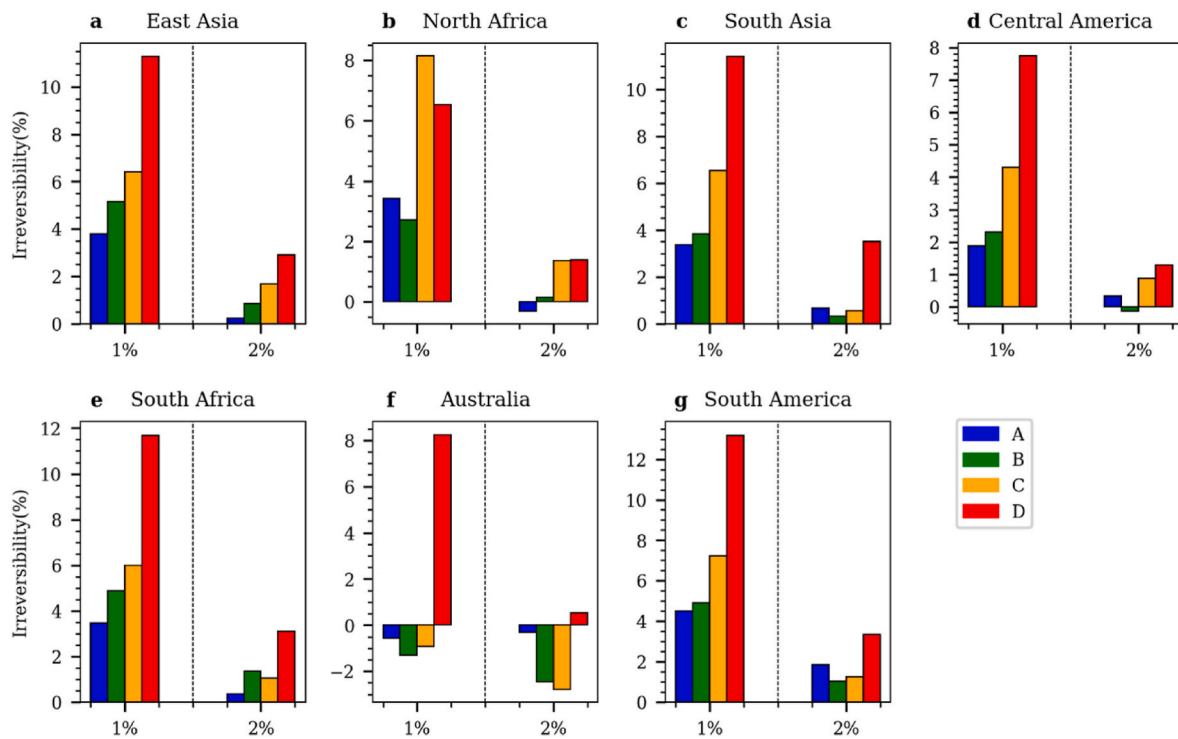


Fig. 3. Quantification of irreversibility for extreme precipitation intensity (Rx1day). (a–g) Irreversibility is quantified as the percentage change in Rx1day between the ramp-down (P_Down) and ramp-up (P_Up) phases, calculated as $((P_Down - P_Up)/P_Up) \times 100$. The x-axis represents increasing delays in the carbon neutrality point, with labels A to D indicating later achievement of carbon neutrality for both 1 % and 2 % CO₂ reduction scenarios. Panels show results for: (a) East Asia, (b) North Africa, (c) South Asia, (d) Central America, (e) South Africa, (f) Australia, and (g) South America.

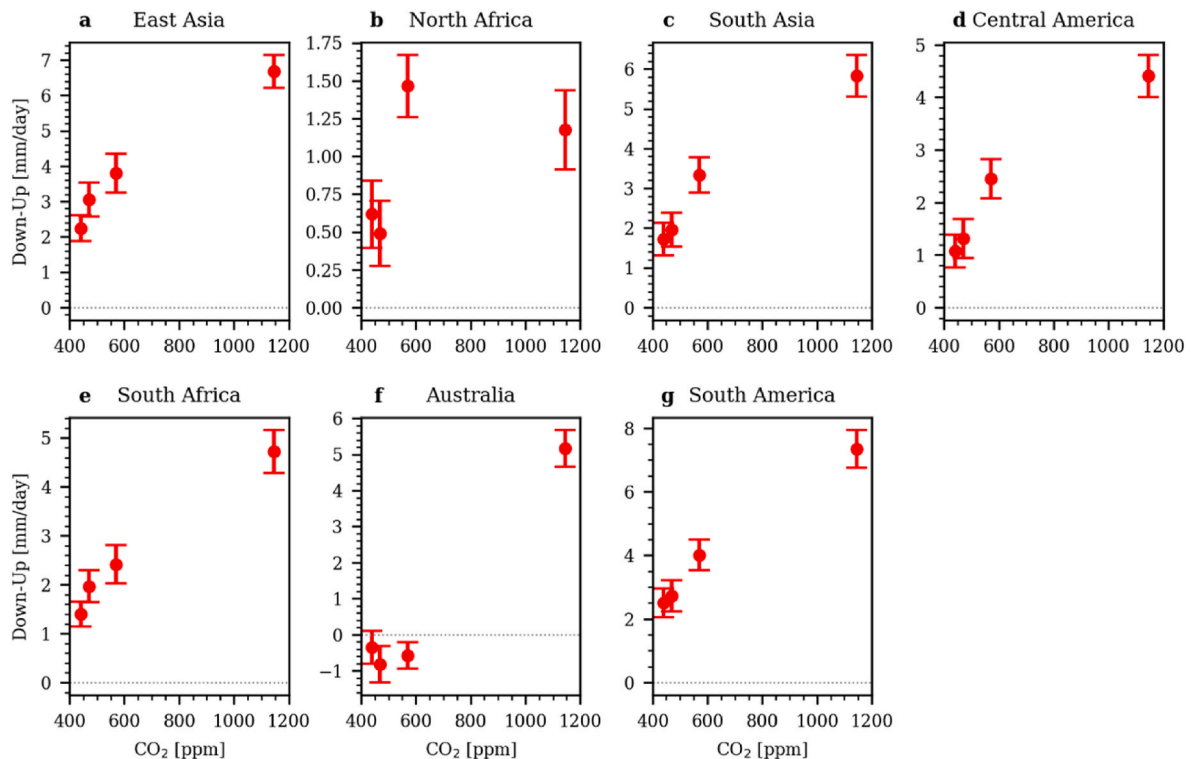


Fig. 4. Regional vulnerability to the timing of carbon neutrality. (a–g) Irreversibility in Rx1day under 1 % per year reduction scenarios across the GMAs: (a) East Asia, (b) North Africa, (c) South Asia, (d) Central America, (e) South Africa, (f) Australia, and (g) South America. The x-axis shows atmospheric CO₂ concentration (in ppm) at the point of carbon neutrality, while the y-axis represents the difference in Rx1day between the ramp-down (P_Down) and ramp-up (P_Up) phases. Red dots indicate the mean, and error bars represent interannual variability.

to return to its initial state following CO₂ reduction and enables to examine regional specific responses according to various timing and rate of carbon neutrality. In the 1 % carbon reduction experiment, the irreversibility of Rx1day intensifies with delays in achieving carbon neutrality, particularly in East Asia, South Asia, Central America, South Africa and South America. These regions exhibit a strong dependency on the timing of peak CO₂ concentrations, with later neutrality timings corresponding to higher degrees of irreversibility. In contrast, North Africa, and Australia show divergent responses. For example, in North Africa, the D1 % scenario exhibits a decrease in Rx1day compared to C1 %, yet it still displays strong irreversible behavior upon carbon reduction, suggesting a more complex hydrological response. In Australia, slight negative irreversibility is observed in the A, B, and C scenarios for both 1 % and 2 % reduction rates, but a strong shift toward positive irreversibility in the D1 % scenario.

The 2 % carbon reduction scenarios consistently resulted in lower levels of irreversibility across all regions compared to the 1 % experiments, even when carbon neutrality was reached at the same concentration levels. This indicates that a faster carbon removal rate can substantially reduce the persistence of extreme precipitation anomalies and mitigate irreversible hydroclimatic responses, implying the importance of timely and aggressive CO₂ mitigation strategies to effectively reduce the persistence of intensified extreme precipitation events across GMAs. Overall, these findings are not sensitive to the choice of different time window length (e.g., 20 or 50 years instead of 30 years), supporting the robustness of the result (Figs. S3 and S4).

The regionally distinct responses of Rx1day to various carbon neutrality forcing offer valuable insight into each region's vulnerability as shaped by the timing of carbon neutrality achievement. To examine this, we analyzed the magnitude of irreversible change, defined as the difference in Rx1day between P_Down and P_Up, as a function of the peak CO₂ concentration at which carbon neutrality attained (Fig. 4). Results show that the magnitude of irreversibility substantially varies by regions. For example, in East Asia, South Asia, South Africa, and Central America, the increase in Rx1day irreversibility with rising peak CO₂ concentrations (or delaying carbon neutrality timing) follows a logarithmic trend, indicating diminishing sensitivity at higher CO₂ levels. In contrast, Australia and South America exhibit an exponential-like increase, where irreversibility accelerates rapidly as peak CO₂ levels rise. North Africa shows a more irregular response pattern, showing higher uncertainty and less consistent trends.

These diverse functional relationships between Rx1day irreversibility and peak CO₂ concentration highlight the region-specific vulnerability to delayed mitigation. In regions exhibiting logarithmic behavior, early carbon mitigation is critical to minimize irreversible impacts, as most of the change occurs at lower CO₂ thresholds. Meanwhile, in regions with exponential responses, the consequences of delayed action become more severe at higher CO₂ levels, suggesting that delayed neutrality could lead to larger impacts. These findings imply the importance of regionally tailored adaptation strategies, with early and aggressive carbon mitigation being especially urgent for areas that respond strongly even at moderate CO₂ levels.

The overall results presented here are based on a single model, NIMS_UKESM1, thus we assess the robustness of Rx1day irreversibility by comparing our findings with a multi-model dataset. Specifically, we use eight Earth system models participating in CMIP6-CDRMIP: ACCESS-ESM1-5, CNRM-ESM2-1, CanESM5, MIROC-ES2L, NorESM2-LM, CESM2, GFDL-ESM4, and UKESM1-0-LL. These models conducted standardized 4 × CO₂ and CO₂-removal experiments following the CDRMIP protocol (Keller et al., 2018), which prescribe a 1 % per year increase in CO₂ concentrations until they reach a quadruple of the pre-industrial level, followed by an equally gradual decrease back to the baseline state. This experimental design closely matches our D1 % experiment, providing a consistent framework for evaluating hysteresis and irreversibility in the global hydrological cycle.

Individual CDRMIP models exhibit large inter-model diversity in the

irreversibility signal. Not all models show a statistically significant difference in extreme precipitation intensity between the P_Down and P_Up phases, and in some regions, the irreversibility even exhibits an opposite sign (Fig. S5). However, when taking the ensemble mean of the CDRMIP models, the results show that, with the exception of North Africa, Rx1day during the P_Down period is intensified relative to the P_Up period in most monsoon regions, consistent with our NIMS_UKESM1 simulation (Fig. 5). These results suggest that the irreversible response of extreme precipitation to delayed CO₂ decline is not model-specific but rather represents a robust and physically consistent feature of the coupled climate system, while highlighting the need for future ensemble-based studies to further constrain structural and parametric uncertainties in projections of extreme precipitation irreversibility.

3.2. Mechanisms driving the irreversibility of extreme precipitation intensity

The intensified extreme precipitation during the P_Down period compared to P_Up can be attributed to increased variability in daily precipitation and/or overall increase in mean precipitation. Both factors contribute to a shift in the precipitation distribution, leading to a higher frequency of extreme precipitation events. When comparing the distributions of daily precipitation between the P_Up and P_Down periods, the differences are statistically significant across all GMAs, as indicated by the Kolmogorov–Smirnov test (Fig. 6). Moreover, each of the seven sub-regions within the GMAs exhibits distinct changes in precipitation patterns. East Asia, South Africa, Australia, and South America, for example, show increases in both mean summer precipitation and its variability during P_Down relative to P_Up, contributing to a higher probability of intense rainfall events that can be described as becoming “wetter and more variable”. This trend is particularly pronounced in East Asia, where the mean daily precipitation increased by ~10 % in P_Down, with a standard deviation increase by ~14 %. Similarly, South Africa, Australia, and South America show moderate increases in mean precipitation, with stronger variability. In these monsoon regions, the irreversibility of mean precipitation under various carbon neutrality scenarios shows patterns similar to those of Rx1day irreversibility (Fig. S6), suggesting that changes in mean precipitation may contribute to explaining the irreversibility of Rx1day.

In contrast, South Asia and Central America exhibit a different response, characterized by increased variability in daily precipitation without a substantial change in the mean. In South Asia, the mean precipitation slightly decreases in mean precipitation in P_Down relative to P_Up, while the standard deviation notably increases from 1.55 to 2.07 mm/day. Central America also experiences a slight increase in mean precipitation, but with enhanced variability.

North Africa stands out as the only region showing a net drying trend during the P_Down period, which can be classified as “drier and more variable”. The mean daily precipitation decreases from 2.76 mm/day in P_Up to 2.67 mm/day in P_Down, while the standard deviation increases from 0.73 to 0.78 mm/day. This combination of reduced overall precipitation and heightened variability suggests a greater risk of hydrological stress in this semi-arid region.

The increased variability of daily precipitation can largely be understood by persistent warming, even after atmospheric CO₂ levels return to their initial levels. This sustained warming is supported by the irreversible response of global SSTs (P_Down – P_Up), with more pronounced warming in scenarios with delayed carbon neutrality (Fig. 7a–d). As shown in previous studies, a warmer atmosphere tends to intensify extreme precipitation, contributing enhanced fluctuations in daily precipitation (O’Gorman and Schneider, 2009; Pendergrass et al., 2017). This thermodynamic effect, driven by increased atmospheric moisture, is a primary driver of enhanced precipitation variability, and aligns with the ~6 % increase in extreme precipitation per °C of warming.

In contrast, changes in mean precipitation, which also influence the

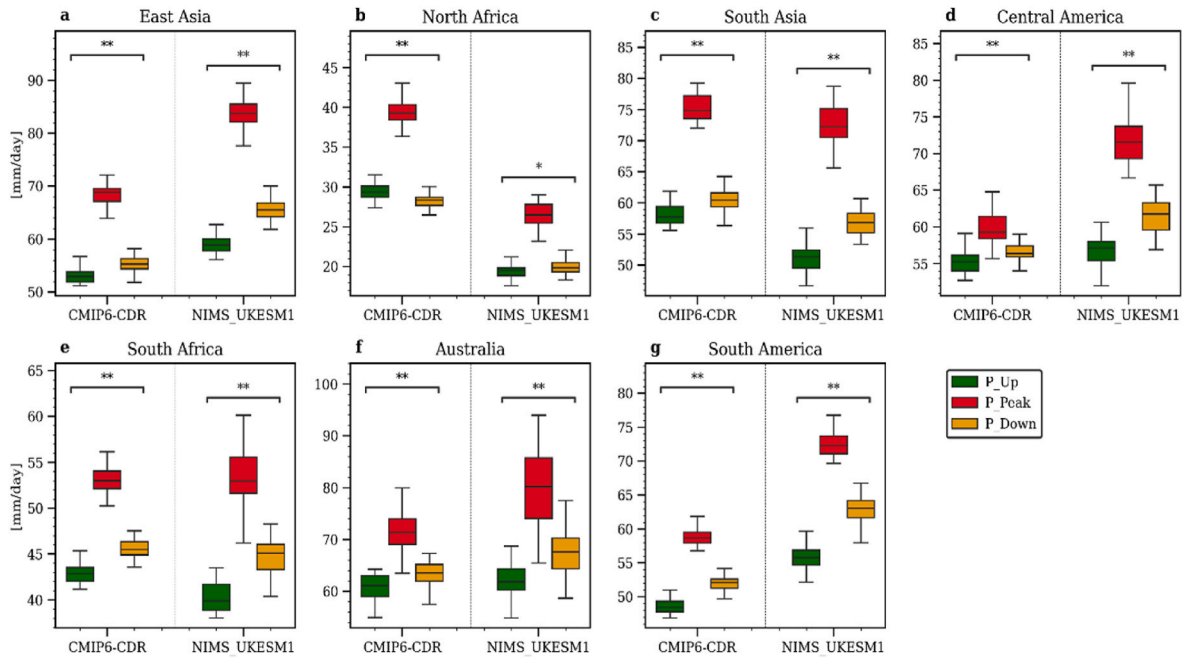


Fig. 5. Comparison of extreme precipitation intensity (Rx1day) across multiple CMIP6-CDR models and the NIMS_UKESM1 simulation under the D1 % carbon neutrality experiment. (a–g) Boxplots show distributions of annual Rx1day values during the ramp-up (P_Up), peak (P_Peak), and ramp-down (P_Down) phases for seven global monsoon regions: (a) East Asia, (b) North Africa, (c) South Asia, (d) Central America, (e) South Africa, (f) Australia, and (g) South America. Green, red, and orange boxes represent the P_Up, P_Peak, and P_Down periods, respectively. Left panels correspond to the CMIP6 multi-model ensemble mean, while right panels show results from the NIMS_UKESM1 simulation. The asterisks indicate statistical significance of differences between the P_Up and P_Down phases based on Welch's *t*-test ($p < 0.05$: *, $p < 0.01$: **). Dashed vertical lines separate CMIP6-CDR and NIMS_UKESM1 results, and whiskers denote the interannual variability range.

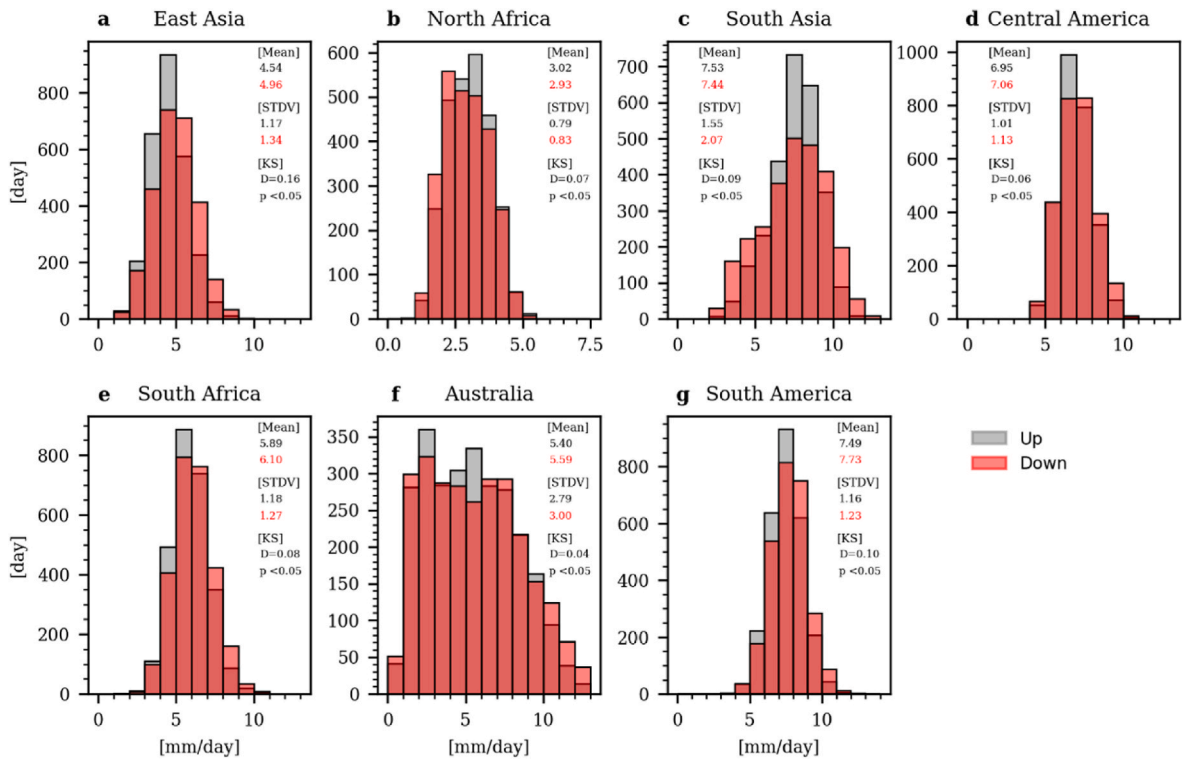


Fig. 6. Distribution of daily precipitation during the P_Up and P_Down phases under the D1 % CO₂ reduction pathway of the carbon neutrality experiment. (a–g) Gray bars represent the histogram of daily precipitation during the P_Up phase, and red bars represent the P_Down phase for each monsoon region: (a) East Asia, (b) North Africa, (c) South Asia, (d) Central America, (e) South Africa, (f) Australia, and (g) South America. Mean and standard deviation values for each phase are provided in each panel (black font for P_Up, red font for P_Down). Results from the Kolmogorov–Smirnov (KS) test are also presented in each panel, showing the test statistic (D) and significance level ($p < 0.05$) to assess whether daily precipitation distributions differ significantly between the P_Up and P_Down.

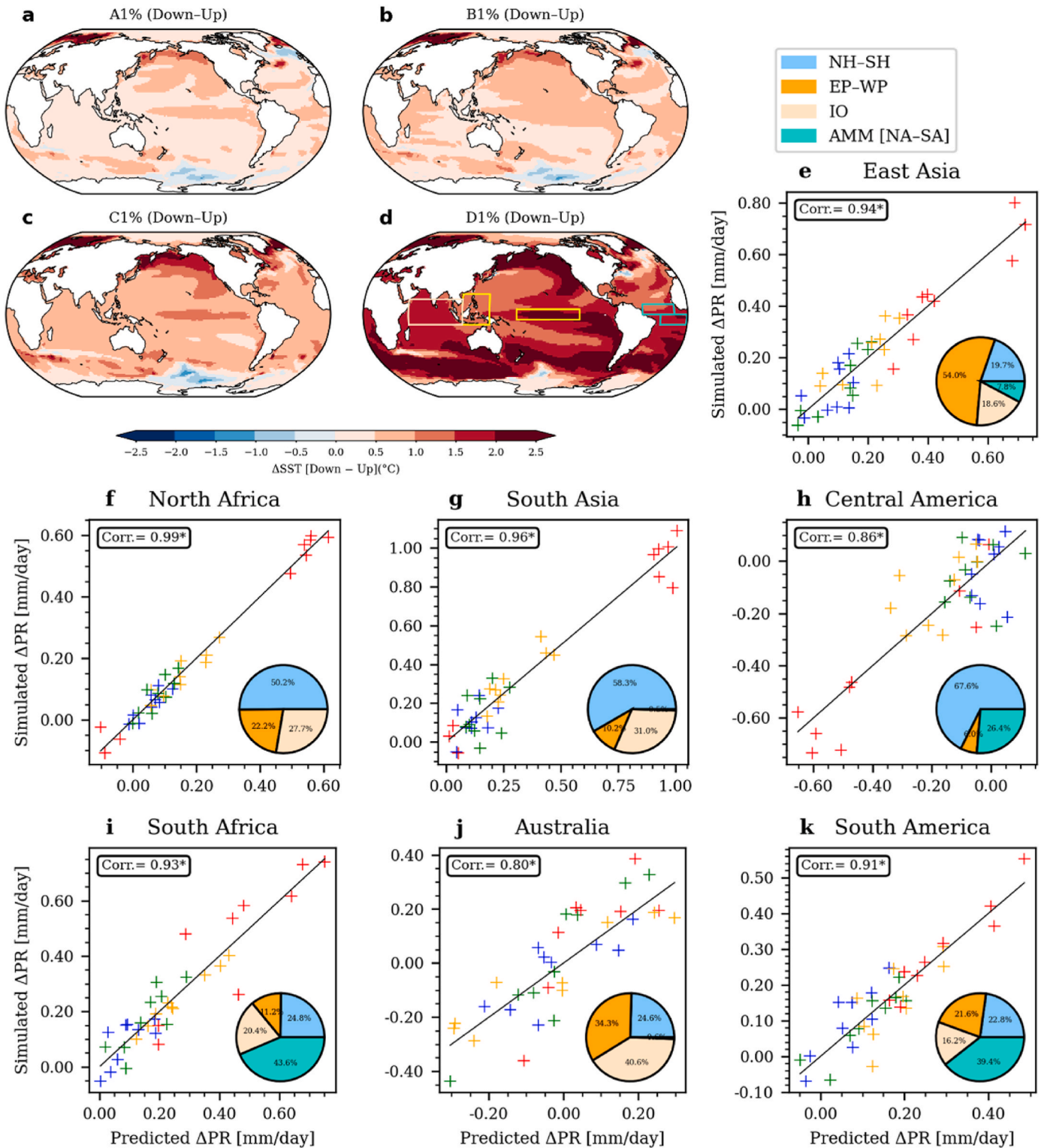


Fig. 7. Irreversibility of global sea surface temperature (SST) under delayed carbon neutrality scenarios and performance of various SST patterns in explaining precipitation irreversibility. (a–d) Global SST changes calculated as the difference between P_Down and P_Up phases under 1 % CO₂ concentration reduction pathways for each carbon neutrality experiment: (a) A1 %, (b) B1 %, (c) C1 %, and (d) D1 %. (e–k) Scatter plots comparing simulated and predicted precipitation changes, using sub-samples from each carbon neutrality experiment. Predicted precipitation changes are obtained from a multi-regression model based on four key SST warming patterns: Global interhemispheric SST gradient (0°–360° E, 0°–60° N minus 0°–360° E, 0°–60° S), Eastern Pacific–Western Pacific gradient (170° E–120° W, 5° S–5° N minus 110° E–140° E, 10° S–20° N), Indian Ocean (50° E–108° E, 10° S–10° N), and Atlantic Meridional Mode (AMM) (50° W–15° W, 0° N–10° N minus 30° W–0° W, 0° S–10° S). Each cross mark, representing one sub-sample, is based on three phase differences (P_Down – P_Up, P_Peak – P_Up, and P_Peak – P_Down) from three ensemble members, resulting in nine cross marks per each carbon neutrality experiment. Blue, green, yellow, and red cross marks represent A1 %, B1 %, C1 %, and D1 % experiments, respectively. Insets (pie charts) within each panel show the relative contributions of the four SST patterns to precipitation changes. All calculations conducted after taking the 30-year running mean.

irreversibility of extreme precipitation, cannot be explained by simple thermodynamic theory alone. Instead, they are strongly modulated by changes in global SST patterns, which influence large-scale atmospheric circulation and, in turn, affect regional precipitation. This is somewhat different from the aforementioned variability changes that are dominated by the increased moisture. Previous studies have demonstrated links between monsoon rainfall and global SST patterns across various monsoon regions (Deng et al., 2018; He et al., 2022; Jiang et al., 2023; Xu et al., 2024; Zhang et al., 2025; Park et al., 2015a,b). Our carbon neutrality experiments similarly show distinct SST warming patterns, including a pronounced interhemispheric warming asymmetry characterized by stronger Northern Hemisphere SST warming compared to the Southern Hemisphere, as well as an El Niño-like warming pattern.

To investigate the extent to which large-scale, persistent SST anomalies account for changes in extreme precipitation over GMAs, we developed a multi regression model that relates precipitation

irreversibility to four key SST warming patterns: the Northern Hemisphere–Southern Hemisphere (NH-SH) SST gradient (0° – 360° E, 0° – 60° N minus 0° – 360° E, 0° – 60° S), the Eastern Pacific–Western Pacific (EP-WP) SST gradient (170° E– 120° W, 5° S– 5° N minus 110° E– 140° E, 10° S– 20° N), the Indian Ocean (IO) SST warming (50° E– 108° E, 10° S– 10° N), and the Atlantic Meridional Mode (AMM) (50° W– 15° W, 0° N– 10° N minus 30° W– 0° W, 0° S– 10° S). These SST patterns explain a large fraction of the difference in precipitation between the P_Down and P_Up phases, accounting for approximately 64–98 % of the variance of precipitation irreversibility (Fig. 7).

However, the dominant SST patterns vary by region (Fig. 7e–k). A variable importance analysis, shown in the inset pie charts within each panel, highlights the relative contributions of the four SST patterns. The EP-WP SST gradient is particularly important for extratropical regions such as East Asia, where the El Niño-like warming pattern dominates the precipitation irreversibility mechanism. In South Africa and South

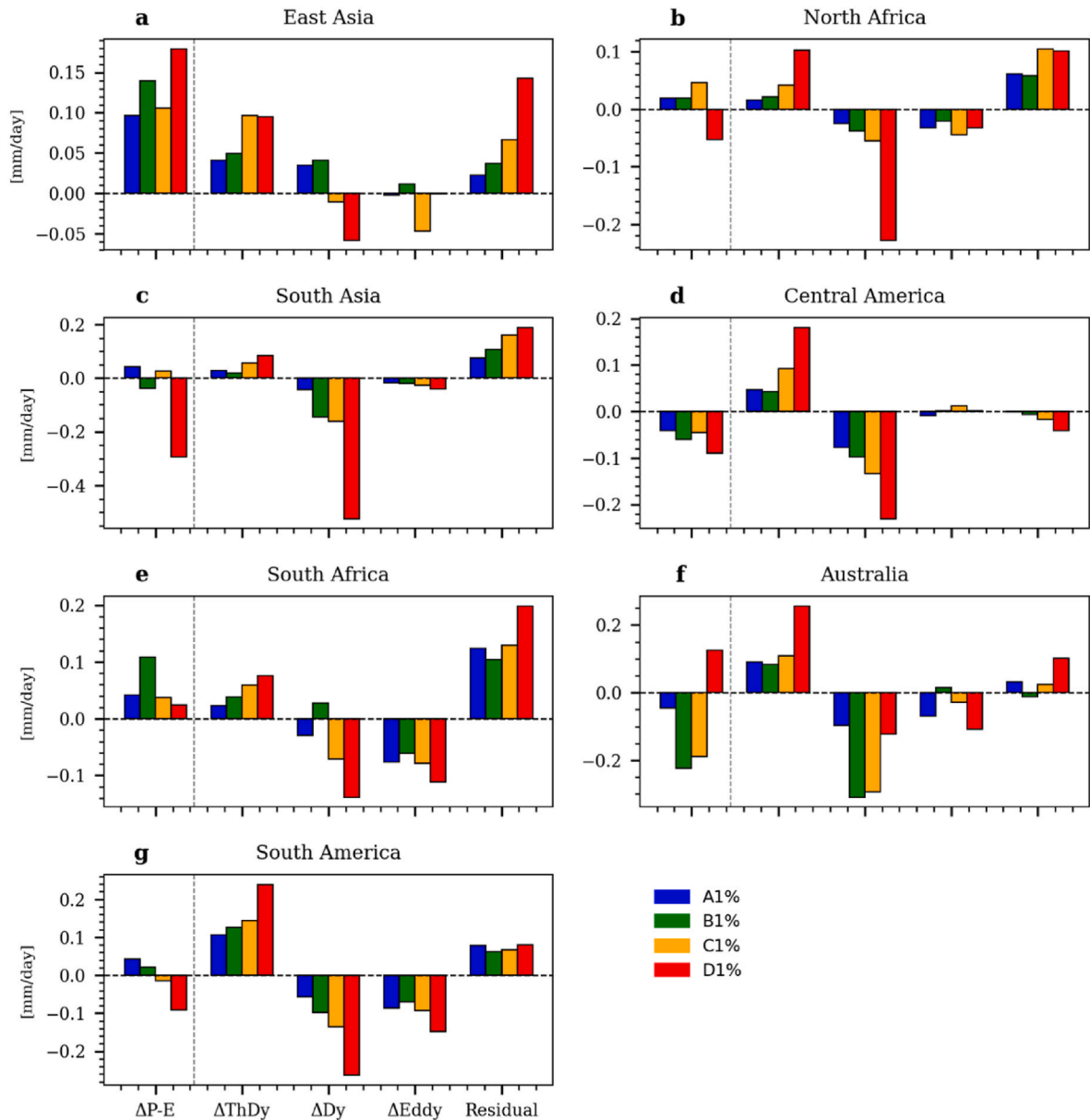


Fig. 8. Changes in the moisture budget of the GMAs under multiple carbon neutrality scenarios. (a–g) Irreversibility of the moisture budget, calculated as the difference between P_Down and P_Up phases under four 1 % per year CO₂ reduction scenarios. The components of moisture budget include ΔP–E (precipitation minus evaporation), ΔThDy (thermodynamic contribution), ΔDy (dynamic contribution), ΔEddy (eddy moisture flux), and Res (residuals). Results are shown for the following monsoon regions: (a) East Asia, (b) North Africa, (c) South Asia, (d) Central America, (e) South Africa, (f) Australia, and (g) South America. All variables are smoothed using a 30-year running mean.

America, the AMM emerges as the primary driver, highlighting the importance of Atlantic SST variability in this region. The NH-SH SST gradient pattern plays a crucial role in tropical regions, particularly in North Africa, South Asia, and Central America, where shifts in the Intertropical Convergence Zone (ITCZ) primarily govern the precipitation response to SST anomalies. These findings imply the critical role of persistent SST anomalies in driving the irreversibility of extreme precipitation, in line with earlier studies (Hatsuzuka and Sato, 2022; Jo et al., 2022; Xi et al., 2015; Zhang et al., 2024).

Although the aforementioned large-scale SST controls governing the regional irreversibility can be interpreted based on previously known linkages, the current results represent only statistical associations. Thus, the explicit physical pathways underlying these relationships under carbon neutrality scenarios need to be further elucidated. To understand a more detailed mechanisms how SST patterns influence the irreversibility of mean precipitation across the GMAs, we conducted a moisture budget analysis comparing changes between the post- and pre-carbon neutrality period (P.Down – P.Up) (Fig. 8). As expected from the regional variations in dominant SST patterns, each monsoon system exhibits distinct contributions from thermodynamic (driven by moisture change) or dynamic (driven by wind change) processes. For example, East Asia shows a clear dominance of thermodynamic contributions to changes in precipitation-minus-evaporation (P-E) (He and Takemi, 2024), with thermodynamic terms exceeding dynamics terms across various carbon neutrality scenarios and following the patterns of P-E changes. This suggests that increased atmospheric moisture driven by higher temperatures over the Indian Ocean and western Pacific, two major moisture sources for the East Asian monsoon, is the primary mechanism driving the irreversibility of extreme precipitation in this extratropical monsoon region.

In contrast, most tropical monsoon regions generally show a stronger influence from dynamic contributions, indicating that changes in atmospheric circulation, rather than moisture availability, play a more substantial role in shaping the irreversibility of mean precipitation. This is presumably consistent with the strong influence of interhemispheric SST gradients, as shown in Fig. 7. The ITCZ shift driven by these SST gradients can modify large-scale circulation patterns, thereby amplifying the role of dynamical processes in tropical monsoon regions. Previous studies have also highlighted the ITCZ shift mechanism as a key factor in understanding precipitation variability and change across tropical monsoon systems (Chiang et al., 2008; Park et al., 2015a,b; Xu et al., 2019).

Although the relative dominance of thermodynamic versus dynamic contributions can broadly explain the precipitation irreversibility in extratropical and tropical monsoon regions, the moisture budget responses remain region-specific, often shaped by the interplay between both processes. For example, in Australia, where multiple SST patterns are equally influential as seen in Fig. 8 dynamic contributions dominate the P-E changes in the earlier carbon neutrality scenarios (A1 %, B1 %, and C1 %). Under the delayed carbon neutrality scenario (D1 %), however, thermodynamic processes become more prominent in the Australian monsoon system. This transition from dynamic to thermodynamic dominance may explain the negative irreversibility in Australia in earlier scenarios and the enhanced irreversibility in D1 %. That is, circulation-driven changes (e.g., a northward shift of the ITCZ induced by higher Northern Hemisphere SST warming relative to Southern Hemisphere) primarily drive the decrease in Rx1day in the earlier carbon neutrality scenarios, whereas increased moisture availability from adjacent warming oceans (e.g., the Indian Ocean and/or western Pacific warming) dominates the Rx1day increase in D1 %. A similar transition is also found in North Africa, where the relative balance between thermodynamic and dynamic factors varies with the timing of carbon neutrality.

4. Conclusions and discussion

This study demonstrates that extreme precipitation intensity across global monsoon areas exhibits irreversible behavior under multiple carbon neutrality scenarios, showing a clear hysteresis in the hydroclimate response. Even when atmospheric CO₂ concentrations return to their initial levels, precipitation extremes fail to revert to their original state. By incorporating multiple combinations of CO₂ peak levels and reduction rates, our carbon neutrality experiments design extends beyond conventional CO₂ ramp-up and ramp-down experiments, which typically employ a single symmetric CO₂ increase up to a quadrupled level followed by an identical decrease, thereby offering only limited insight into the heterogeneous responses of hydroclimatic extremes. Our multi-pathway approach reveals that the degree of irreversibility is substantially affected by both the timing of carbon neutrality and the rate of CO₂ reduction, with delayed and slower reduction amplifying hysteresis effects. These findings highlight that the timing of carbon neutrality is as critical as the ultimate emission target in determining the persistence of extreme precipitation. The multi-pathway design thus offers a novel framework for diagnosing the climatic implications of delayed or accelerated mitigation and for identifying regional thresholds of hydroclimatic irreversibility.

Pronounced regional variations are also evident, with distinct responses and underlying mechanisms across different monsoon regions. Such irreversible changes are driven by a combination of **thermodynamic factors**, linked to increased atmospheric moisture, and **dynamic processes**, associated with shifts in large-scale atmospheric circulation. Persistent sea surface temperature anomalies, particularly those characterized by Pacific zonal SST gradients and global/regional interhemispheric SST differences play a critical role in sustaining the region-specific disruptions.

While the finding that delayed and slower CO₂ reductions increase irreversibility may be anticipated, a major insight of this study is the pronounced regional heterogeneity in the behavior of irreversible responses to different timing of carbon neutrality. In particularly sensitive regions, such as East Asia and South Africa, even slight delays in achieving carbon neutrality can lead to a steep increase in hydroclimatic irreversibility, whereas in less sensitive regions such as Australia and South America, a comparable increase occurs only under later carbon neutrality timings. The heightened sensitive in East Asian region can be attributed to enhanced moisture flux by intensified warming over the Indian Ocean and western Pacific, reflecting the dominant thermodynamic influence in this region. Similarly, the delayed sensitivity of precipitation irreversibility in Australia can be understood through its transition from dominant dynamical control under earlier carbon neutrality timings to dominant thermodynamic control under later timings. This pattern reflects a less sensitive response of Australian precipitation to ITCZ shifts driven by interhemispheric differential warming in the earlier scenarios and a stronger response to moisture supply from adjacent oceans in the later ones. Nevertheless, delayed mitigation not only prolongs the duration of extreme conditions but increases the likelihood of crossing regional tipping points, defined here as situations where the system fails to return to its initial state within a human generational timescale.

This has critical implications for climate adaptation. Regional strategies need to account for the possibility of locked-in hydroclimatic extremes, including heightened frequency and severity of flooding. Incorporating these projections into long-term water resource planning, resilient infrastructure design, and disaster risk management is crucial, especially for monsoon-dependent economies with high exposure and vulnerability. For example, in South Asia and East Asia, flooding is very impactful phenomena (Kimuli et al., 2021; Monioudi et al., 2025), thus adaptation planning should be focused on flood management, given the strong increase in irreversibility associated with even small delays in achieving carbon neutrality. In Australia, where alternating wet and dry extremes emerge from compensating thermodynamic and dynamic

effects (Li et al., 2021; Mohanty et al., 2024), enhancing water storage capacity and irrigation resilience will be key components of adaptation planning. In other regions, such as Central and South America, flexible reservoir operations and the maintenance of forest cover and moisture recycling are essential to reducing hydroclimatic instability under delayed mitigation scenarios (Ley et al., 2023; Ruv Lemes et al., 2023).

The model employed in this study does not account for internal carbon-climate feedbacks, such as land-atmosphere interactions (e.g., land-use change, deforestation, irrigation) and ocean-atmosphere processes (e.g., phytoplankton-driven changes in carbon uptake and heat flux). This feedback can influence both atmospheric CO₂ concentrations and the surface energy budget, and may substantially modulate or amplify climate responses in monsoon-dominated regions where land and ocean dynamics are tightly coupled (Ilyina et al., 2021; Park et al., 2015a,b; Pielke et al., 2002). Future modeling efforts aimed at understanding the irreversible responses of extreme precipitation should therefore aim to integrate these complex feedback mechanisms to better capture the full spectrum of potential climate trajectories. This integration will enhance the realism of hydroclimate projections and support the development of more robust, region-specific mitigation and adaptation strategies for vulnerable monsoon systems.

CRedit authorship contribution statement

Md. Babul Miah: Writing – original draft, Visualization, Methodology. **Jong-Yeon Park:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization. **Min-Uk Lee:** Validation, Data curation. **Woojin Jeon:** Writing – review & editing, Validation. **Young-Hwa Byun:** Supervision, Conceptualization. **Hyun Min Sung:** Methodology, Conceptualization. **Jin Gi Hong:** Writing – review & editing. **Md. Jalal Uddin:** Writing – review & editing. **Sanjit Kumar Mondal:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.wace.2025.100843>.

Data availability

The analysis codes and multiple carbon neutrality experiment datasets generated using the UKESM1 model are archived in Zenodo [Miah et al., 2025].

References

- Chadwick, R., Wu, P., Good, P., Andrews, T., 2013. Asymmetries in tropical rainfall and circulation patterns in idealised CO₂ removal experiments. *Clim. Dyn.* 40 (1–2). <https://doi.org/10.1007/s00382-012-1287-2>.
- Chiang, J.C.H., Vimont, D.J., 2004. Analogous Pacific and Atlantic meridional modes of tropical atmosphere-ocean variability. *J. Clim.* 17 (21). <https://doi.org/10.1175/JCLI4953.1>.
- Chiang, J.C.H., Fang, Y., Chang, P., 2008. Interhemispheric thermal gradient and tropical Pacific climate. *Geophys. Res. Lett.* 35 (14). <https://doi.org/10.1029/2008GL034166>.

- Chuphal, D.S., Malik, I., Singh, R., Vangala, G., Niranjanika, M., Vegad, U., et al., 2025. Multi-day extreme precipitation caused major floods in India during summer monsoon of 2024. *Earth's Future* 13 (5). <https://doi.org/10.1029/2024EF005497>.
- Deng, K., Yang, S., Ting, M., Tan, Y., He, S., 2018. Global monsoon precipitation: trends, leading modes, and associated drought and heat wave in the Northern hemisphere. *J. Clim.* 31 (17). <https://doi.org/10.1175/JCLI-D-17-0569.1>.
- Hatsuzuka, D., Sato, T., 2022. Impact of SST on present and future extreme precipitation in Hokkaido investigated considering weather patterns. *J. Geophys. Res. Atmos.* 127 (9). <https://doi.org/10.1029/2021JD036120>.
- He, P., Liu, J., Wang, B., Sun, W., 2022. Understanding global monsoon precipitation changes during the 8.2 ka event and the current warm period. *Palaeogeogr. Palaeoclimatol. Palaeoecol.* 586. <https://doi.org/10.1016/j.palaeo.2021.110757>.
- He, S., Takemi, T., 2024. Future changes of extreme precipitation and related atmospheric conditions in east Asia under global warming projected in large ensemble climate prediction data. *J. Clim.* 37 (19), 5171–5186. <https://doi.org/10.1175/JCLI-D-22-0924.1>.
- Hirabayashi, Y., Mahendran, R., Koirala, S., Konoshima, L., Yamazaki, D., Watanabe, S., et al., 2013. Global flood risk under climate change. *Nat. Clim. Change* 3 (9). <https://doi.org/10.1038/nclimate1911>.
- Hwang, J., Son, S.W., Garfinkel, C.I., Woollings, T., Yoon, H., An, S. Il, et al., 2024. Asymmetric hysteresis response of mid-latitude storm tracks to CO₂ removal. *Nat. Clim. Change* 14 (5). <https://doi.org/10.1038/s41558-024-01971-x>.
- Ilyina, T., Li, H., Spring, A., Müller, W.A., Bopp, L., Chikamoto, M.O., et al., 2021. Predictable variations of the carbon sinks and atmospheric CO₂ growth in a multi-model framework. *Geophys. Res. Lett.* 48 (6). <https://doi.org/10.1029/2020GL090695>.
- Im, N., Kim, D., An, S. Il, Paik, S., Kim, S.K., Shin, J., et al., 2024. Hysteresis of european summer precipitation under a symmetric CO₂ ramp-up and ramp-down pathway. *Environ. Res. Lett.* 19 (7). <https://doi.org/10.1088/1748-9326/ad52ad>.
- IPCC, 2018. IPCC, 2018: Global warming of 1.5°C. An IPCC special report on the impacts of global warming of 1.5°C above pre-industrial levels and related global greenhouse gas emission pathways. In: The Context of Strengthening the Global Response to the Threat of Climate Change, Sustainable Development, and Efforts to Eradicate Poverty. Cambridge University Press. Cambridge University Press. <https://doi.org/10.1017/9781009157940>.
- IPCC, 2021. Technical summary. In: Climate Change 2021: the Physical Science Basis. Contribution of Working Group I to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change. Cambridge University Press. <https://doi.org/10.1017/9781009157896.002>.
- IPCC, 2023. Climate change 2021 : the physical science basis : Working group I contribution to the sixth assessment report of the intergovernmental panel on climate change. <https://doi.org/10.1017/9781009157889>.
- Jiang, Y., Li, J., Wang, B., Yang, Y., Zhu, Z., 2023. Weakening of decadal variation of Northern hemisphere land monsoon rainfall under global warming. *npj Clim. Atmos. Sci.* 6 (1). <https://doi.org/10.1038/s41612-023-00441-z>.
- Jo, S.Y., Seong, M.G., Min, S.K., Kug, J.S., Yeh, S.W., An, S. Il, et al., 2022. Hysteresis behaviors in East Asian extreme precipitation frequency to CO₂ pathway. *Geophys. Res. Lett.* 49 (18). <https://doi.org/10.1029/2022GL099814>.
- Joos, F., Spahni, R., 2008. Rates of change in natural and anthropogenic radiative forcing over the past 20,000 years. *Proc. Natl. Acad. Sci. USA* 105 (5). <https://doi.org/10.1073/pnas.0707386105>.
- Keller, D.P., Lenton, A., Scott, V., Vaughan, N.E., Bauer, N., Ji, D., et al., 2018. The Carbon Dioxide Removal Model Intercomparison Project (CDRMIIP): rationale and experimental protocol for CMIP6. *Geosci. Model Dev. (GMD)* 11 (3). <https://doi.org/10.5194/gmd-11-1133-2018>.
- Kim, S.K., Shin, J., An, S. Il, Kim, H.J., Im, N., Xie, S.P., et al., 2022. Widespread irreversible changes in surface temperature and precipitation in response to CO₂ forcing. *Nat. Clim. Change* 12 (9). <https://doi.org/10.1038/s41558-022-01452-z>.
- Kimuli, J.B., Di, B., Zhang, R., Wu, S., Li, J., Yin, W., 2021. A multisource trend analysis of floods in Asia-Pacific 1990–2018: implications for climate change in sustainable development goals. *Int. J. Disaster Risk Reduct.* <https://doi.org/10.1016/j.ijdrr.2021.102237>.
- Kriegler, E., Luderer, G., Bauer, N., Baumstark, L., Fujimori, S., Popp, A., et al., 2018. Pathways limiting warming to 1.5°C: a tale of turning around in no time? *Philos. Trans. R. Soc. A Math. Phys. Eng. Sci.* 376 (2119). <https://doi.org/10.1098/rsta.2016.0457>.
- Kug, J.S., Oh, J.H., An, S. Il, Yeh, S.W., Min, S.K., Son, S.W., et al., 2022. Hysteresis of the intertropical convergence zone to CO₂ forcing. *Nat. Clim. Change* 12 (1), 47–53. <https://doi.org/10.1038/s41558-021-01211-6>.
- Kushnir, Y., Robinson, W.A., Bladé, I., Hall, N.M.J., Peng, S., Sutton, R., 2002. Atmospheric GCM response to extratropical SST anomalies: synthesis and evaluation. *J. Clim.* 15 (16). [https://doi.org/10.1175/1520-0442\(2002\)015<2233:AGRTES>2.0.CO;2](https://doi.org/10.1175/1520-0442(2002)015<2233:AGRTES>2.0.CO;2).
- Ley, D., Guillén Bolaños, T., Castaneda, A., Hidalgo, H.G., Girot Pignot, P.O., Fernández, R., et al., 2023. Central America urgently needs to reduce the growing adaptation gap to climate change. *Front. Clim.* <https://doi.org/10.3389/fclim.2023.1215062>.
- Li, Y., Zhang, Y., Ye, W., Zhang, X., 2021. Global wet/dry patterns and mechanisms since the last glacial maximum: a key to future projection. *Earth's Future* 9 (10). <https://doi.org/10.1029/2020EF001907>.
- Liu, C., An, S. Il, Jin, F.F., Shin, J., Kug, J.S., Zhang, W., et al., 2023. Hysteresis of the El Niño–Southern oscillation to CO₂ forcing. *Sci. Adv.* 9 (31). <https://doi.org/10.1126/sciadv.adh8442>.
- McKay, D.I.A., Staal, A., Abrams, J.F., Winkelmann, R., Sakschewski, B., Loriani, S., et al., 2022. Exceeding 1.5°C global warming could trigger multiple climate tipping points. *Science* 377 (6611). <https://doi.org/10.1126/science.abn7950>.

- Miah, M.B., Park, J.-Y., Lee, M.-U., Jeon, W., Byun, Y.-H., Sung, H.-M., Hong, J.G., Uddin, M.J., Mondal, S.K., 2025. Code and datasets for "Irreversibility of extreme precipitation intensity in global monsoon areas under multiple carbon neutrality scenarios". Zenodo. <https://doi.org/10.5281/zenodo.17063562>.
- Mohanty, S., Jakob, C., Singh, M.S., 2024. Australian summer monsoon bursts: a moist static energy budget perspective. *J. Geophys. Res. Atmos.* 129 (1). <https://doi.org/10.1029/2023JD039048>.
- Mondal, S.K., An, S. Il, Min, S.K., Kim, S.K., Shin, J., Paik, S., et al., 2023. Hysteresis and irreversibility of global extreme precipitation to anthropogenic CO₂ emission. *Weather Clim. Extrem.* 40. <https://doi.org/10.1016/j.wace.2023.100561>.
- Monioudi, I.N., Voudoukas, M.I., Giardino, A., Stocchino, A., Mentaschi, L., Feyen, L., 2025. Impacts of sea level rise and adaptation across Asia and the Pacific. *Sci. Rep.* 15 (1), 35742. <https://doi.org/10.1038/s41598-025-11517-6>.
- O'Gorman, P.A., Schneider, T., 2009. The physical basis for increases in precipitation extremes in simulations of 21st-century climate change. *Proc. Natl. Acad. Sci. USA* 106 (35). <https://doi.org/10.1073/pnas.0907610106>.
- Oh, H., An, S. Il, Shin, J., Yeh, S.W., Min, S.K., Son, S.W., Kug, J.S., 2022. Contrasting hysteresis behaviors of Northern hemisphere land monsoon precipitation to CO₂ pathways. *Earth's Future* 10 (6). <https://doi.org/10.1029/2021EF002623>.
- Otto, F.E.L., Zachariah, M., Saeed, F., Siddiqi, A., Kamil, S., Mushtaq, H., et al., 2023. Climate change increased extreme monsoon rainfall, flooding highly vulnerable communities in Pakistan. *Environ. Res.: Climate* 2 (2). <https://doi.org/10.1088/2752-5295/acbfd5>.
- Paik, S., An, S. Il, Min, S.K., King, A.D., Shin, J., 2023. Hysteretic behavior of global to regional monsoon area under CO₂ Ramp-Up and ramp-down. *Earth's Future* 11 (7). <https://doi.org/10.1029/2022EF003434>.
- Park, J.Y., Kug, J.S., Bader, J., Rolph, R., Kwon, M., 2015a. Amplified Arctic warming by phytoplankton under greenhouse warming. *Proc. Natl. Acad. Sci. USA* 112 (19). <https://doi.org/10.1073/pnas.1416884112>.
- Park, J.Y., Bader, J., Matei, D., 2015b. Northern-hemispheric differential warming is the key to understanding the discrepancies in the projected Sahel rainfall. *Nat. Commun.* 6. <https://doi.org/10.1038/ncomms6985>.
- Pendergrass, A.G., Knutti, R., Lehner, F., Deser, C., Sanderson, B.M., 2017. Precipitation variability increases in a warmer climate. *Sci. Rep.* 7 (1). <https://doi.org/10.1038/s41598-017-17966-y>.
- Pielke, R.A., Marland, G., Betts, R.A., Chase, T.N., Eastman, J.L., Niles, J.O., et al., 2002. The influence of land-use change and landscape dynamics on the climate system: relevance to climate-change policy beyond the radiative effect of greenhouse gases. *Philos. Trans. R. Soc. A Math. Phys. Eng. Sci.* <https://doi.org/10.1098/rsta.2002.1027>.
- Pomposi, C., Kushnir, Y., Giannini, A., 2015. Moisture budget analysis of SST-driven decadal Sahel precipitation variability in the twentieth century. *Clim. Dyn.* 44 (11–12). <https://doi.org/10.1007/s00382-014-2382-3>.
- Quéré, C., Andrew, R., Friedlingstein, P., Sitch, S., Hauck, J., Pongratz, J., et al., 2018. Global carbon budget 2018. *Earth Syst. Sci. Data* 10 (4), 2141–2194. <https://doi.org/10.5194/essd-10-2141-2018>.
- Ruv Lemes, M., Sampaio, G., Garcia-Carreras, L., Fisch, G., Alves, L.M., Bassett, R., et al., 2023. Impacts on South America moisture transport under Amazon deforestation and 2 °C global warming. *Sci. Total Environ.* 905. <https://doi.org/10.1016/j.scitotenv.2023.167407>.
- Santana-Falcón, Y., Yamamoto, A., Lenton, A., Jones, C.D., Burger, F.A., John, J.G., et al., 2023. Irreversible loss in marine ecosystem habitability after a temperature overshoot. *Commun. Earth Environ.* 4 (1). <https://doi.org/10.1038/s43247-023-01002-1>.
- Sellar, A.A., Jones, C.G., Mulcahy, J.P., Tang, Y., Yool, A., Wiltshire, A., et al., 2019. UKESM1: description and evaluation of the U.K. Earth system model. *J. Adv. Model. Earth Syst.* 11 (12). <https://doi.org/10.1029/2019MS001739>.
- Senior, C.A., Jones, C.G., Wood, R.A., Sellar, A., Belcher, S., Klein-Tank, A., et al., 2020. U.K. community Earth system modeling for CMIP6. *J. Adv. Model. Earth Syst.* 12 (9). <https://doi.org/10.1029/2019MS002004>.
- Shi, X., Chen, J., Gu, L., Xu, C.Y., Chen, H., Zhang, L., 2021. Impacts and socioeconomic exposures of global extreme precipitation events in 1.5 and 2.0 °C warmer climates. *Sci. Total Environ.* 766. <https://doi.org/10.1016/j.scitotenv.2020.142665>.
- Solomon, S., Plattner, G.K., Knutti, R., Friedlingstein, P., 2009. Irreversible climate change due to carbon dioxide emissions. *Proc. Natl. Acad. Sci. USA* 106 (6). <https://doi.org/10.1073/pnas.0812721106>.
- Su, X., Wang, L., Huang, G., Wang, T., Piao, J., Wang, Q., 2025. Summer precipitation responses to CO₂ removal scenario over the transitional climate zone in East Asia and the driving mechanisms. *Clim. Dyn.* 63 (3). <https://doi.org/10.1007/s00382-025-07647-2>.
- Trenberth, K.E., Stepaniak, D.P., Caron, J.M., 2000. The global monsoon as seen through the divergent atmospheric circulation. *J. Clim.* 13 (22), 2. [https://doi.org/10.1175/1520-0442\(2000\)013<3969:TGMAS7>2.0.CO](https://doi.org/10.1175/1520-0442(2000)013<3969:TGMAS7>2.0.CO).
- Turner, A.G., Slingo, J.M., 2009. Uncertainties in future projections of extreme precipitation in the Indian monsoon region. *Atmos. Sci. Lett.* 10 (3). <https://doi.org/10.1002/asl.223>.
- Vaughan, N.E., Lenton, T.M., 2011. A review of climate geoengineering proposals. *Clim. Change.* <https://doi.org/10.1007/s10584-011-0027-7>.
- Wang, B., Ding, Q., 2006. Changes in global monsoon precipitation over the past 56 years. *Geophys. Res. Lett.* 33 (6). <https://doi.org/10.1029/2005GL025347>.
- Wang, B., Ding, Q., 2008. Global monsoon: dominant mode of annual variation in the tropics. *Dynam. Atmos. Oceans* 44 (3–4), 165–183. <https://doi.org/10.1016/j.dynatmoce.2007.05.002>.
- Wang, B., Biasutti, M., Byrne, M.P., Castro, C., Chang, C.P., Cook, K., et al., 2021. Monsoons climate change assessment. *Bull. Am. Meteorol. Soc.* 102 (1), E1–E19. <https://doi.org/10.1175/BAMS-D-19-0335.1>.
- Winsemius, H.C., Aerts, J.C.J.H., Van Beek, L.P.H., Bierkens, M.F.P., Bouwman, A., Jongman, B., et al., 2016. Global drivers of future river flood risk. *Nat. Clim. Change* 6 (4). <https://doi.org/10.1038/nclimate2893>.
- World Meteorological Organization (WMO), 2017. WMO Guidelines on the Calculation of Climate Normals, 978–92–63–11203–3. WMO, Geneva. Retrieved from: <https://library.wmo.int/idurl/4/55797>.
- Wu, P., Wood, R., Ridley, J., Lowe, J., 2010. Temporary acceleration of the hydrological cycle in response to a CO₂ rampdown. *Geophys. Res. Lett.* 37 (12). <https://doi.org/10.1029/2010GL043730>.
- Xi, J., Zhou, L., Murtugudde, R., Jiang, L., 2015. Impacts of intraseasonal SST anomalies on precipitation during Indian summer monsoon. *J. Clim.* 28 (11). <https://doi.org/10.1175/JCLI-D-14-00096.1>.
- Xie, S.P., Carton, J.A., 2004. Tropical Atlantic variability: patterns, mechanisms, and impacts. In: *Geophysical Monograph Series*, vol. 147. <https://doi.org/10.1029/147GM07>.
- Xu, Hai, Song, Y., Goldsmith, Y., Lang, Y., 2019. Meridional ITCZ shifts modulate tropical/subtropical Asian monsoon rainfall. *Sci. Bull.* <https://doi.org/10.1016/j.scib.2019.09.025>.
- Xu, Huiwen, Chen, H., Wang, H., 2022. Increased populations will be exposed to the dangerous precipitation extremes across China in the future. *Front. Earth Sci.* 10. <https://doi.org/10.3389/feart.2022.963042>.
- Xu, Z., Kosaka, Y., Toda, M., Iwakiri, T., Huang, G., Ji, F., et al., 2024. Irreversibility of winter precipitation over the Northeastern Pacific and Western North America against CO₂ forcing. *npj Clim. Atmos. Sci.* 7 (1). <https://doi.org/10.1038/s41612-024-00864-2>.
- Yool, A., Palmieri, J., Jones, C.G., Sellar, A.A., de Mora, L., Kuhlbrodt, T., et al., 2020. Spin-up of UK Earth system model 1 (UKESM1) for CMIP6. *J. Adv. Model. Earth Syst.* 12 (8). <https://doi.org/10.1029/2019MS001933>.
- Zhang, L., Lin, J., Chen, Z., Shi, Y., Du, Y., 2025. Quantifying impacts of two ENSO patterns on South China sea summer monsoon onset. *Geosci. Lett.* 12 (1). <https://doi.org/10.1186/s40562-025-00376-8>.
- Zhang, W., Tao, W., Huang, G., Hu, K., Qu, X., Gong, H., et al., 2024. Irreversibility of ENSO impacts on the wintertime anomalous Western North Pacific anticyclone to CO₂ forcing. *npj Clim. Atmos. Sci.* 7 (1). <https://doi.org/10.1038/s41612-024-00854-4>.
- Zhang, X., Alexander, L., Hegerl, G.C., Jones, P., Tank, A.K., Peterson, T.C., et al., 2011. Indices for monitoring changes in extremes based on daily temperature and precipitation data. *Wiley Interdiscip. Rev.: Clim. Change.* <https://doi.org/10.1002/wcc.147>.
- Zhou, T., Zhang, L., Li, H., 2008. Changes in global land monsoon area and total rainfall accumulation over the last half century. *Geophys. Res. Lett.* 35 (16). <https://doi.org/10.1029/2008GL034881>.
- Zickfeld, K., Eby, M., Weaver, A.J., Alexander, K., Cresspin, E., Edwards, N.R., et al., 2013. Long-term climate change commitment and reversibility: an EMIC intercomparison. *J. Clim.* 26 (16). <https://doi.org/10.1175/JCLI-D-12-00584.1>.
- Zitoun, R., Marcinek, S., Hatje, V., Sander, S.G., Völker, C., Sarin, M., Omanović, D., 2024. Climate change driven effects on transport, fate and biogeochemistry of trace element contaminants in coastal marine ecosystems. *Commun. Earth Environ.* <https://doi.org/10.1038/s43247-024-01679-y>.